solutions of this inequation are 'natural numbers'. Now,

Add 3 on both sides, we get $3x - 3 + 3 < 12 + 3 \Rightarrow \quad 3x < 15$

Divide by 3 on both sides, we get $\frac{3x}{3} < \frac{15}{3} \Rightarrow x < 5$

Hence, x takes value which is less than 5 and x is a natural number. Thus, the solution for this inequation are 1, 2, 3 and 4.

Note

When 'x' is not restricted to a natural number, the solution includes all values less than 5.

Rules to solve inequation

While solving an inequation, the rules for transposition in case of inequalities are the same as for equations.

1. **Addition** of the same number on both sides of the inequation does not change the value of the inequation. Example: $10 > 5 \Rightarrow 10 + \mathbf{1} > 5 + \mathbf{1} \Rightarrow 11 > 6$.

   Extending this result, when adding any number 'x' instead of 1, the inequality $10 + x > 5 + x$ remains unchanged.

2. **Subtraction** of the same number from both sides of the inequation does not change the value of the inequation. Example: $10 > 5 \Rightarrow 10 - \mathbf{1} > 5 - \mathbf{1} \Rightarrow 9 > 4$.

   Extending this result, when subtracting any number x instead of 1, the inequatity $10 - x > 5 - x$ remains unchanged.

3. **Multiplication** by the same *positive number* on both sides of the inequation does not change the value of the inequation. Example: $10 > 5 \Rightarrow 10 \times \mathbf{2} > 5 \times \mathbf{2} \Rightarrow 20 > 10$.

   Similarly, when multiplying any positive number x instead of 2, the inequatity $10 \times x > 5 \times x$ remains unchanged.

4. **Division** by the same *non-zero positive number* on both sides of the inequation does not change the value of the inequation. Example: $10 > 5 \Rightarrow \frac{10}{5} > \frac{5}{5} \Rightarrow 2 > 1$.

   Similarly, when dividing any non zero positive number x instead of 5, the inequatity $\frac{10}{x} > \frac{5}{x}$ remains unchanged.

Note

When both sides of an inequation are multiplied or divided by the same *non-zero negative number*, the sign of inequality is reversed. For example, consider $3 < 12$.

Multiplying -1 on both sides, we get, $3 \times (-1) < 12 \times (-1)$

$$-3 < -12.$$

But, it is not true. Because, -3 is greater than -12.

So, $-3 > -12$. Note that the sign of inequality is reversed.

To generalize , when $x < y$ is multiplied by -1 on both sides , we get $-x > -y$.

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